Christina Myalla

Research Data Scientist | Data Engineer | Geospatial Health Analyst
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PROFESSIONAL SUMMARY

Research Data Scientist and Data Engineer with over 10 years of experience spanning health research, geospatial analysis, data systems development, and research capacity strengthening across Tanzania and the broader sub-Saharan Africa region. Currently contributing to the Malaria Atlas Project (MAP) at Ifakara Health Institute and The Kids Research Institute Australia, where I lead the Dar es Salaam Node's Data Engineering Team, with a focus on building ETL pipelines for multi-country health facility datasets, geospatial accessibility modelling, and developing interactive data products for National Malaria Control Programmes. Holds a Master of Science in Health Informatics (University of Dar es Salaam, 2025) and a Bachelor of Science in Information Systems (University of Dodoma). Combines deep technical expertise in R, Python, GIS, and data engineering with a strong commitment to translating complex analyses into actionable evidence for public health system planning in low- and middle-income country settings.

WORK EXPERIENCE

Research Data Scientist April 2025 to Present
Malaria Atlas Project (MAP), Dar es Salaam Node, Ifakara Health Institute Dar es Salaam, Tanzania / Perth, Australia

- Lead the Node's Data Engineering Team (DET), coordinating data pipelines, tools and platforms, and mentoring junior team members.
- Design and implement ETL (Extract-Transform-Load) data pipelines to harmonize multi-country health facility datasets, including demographics, geolocation, and service availability, enabling high-quality inputs for malaria modelling and health systems analysis across sub-Saharan Africa.
- Develop and deploy interactive data products and visualization tools, including Shiny dashboards and Quarto-based portals, to support evidence-based decision-making by National Malaria Control Programmes (NMCPs) and partner organizations.
- Conduct advanced geospatial and accessibility analyses, including travel-time modelling using the MAP routing engine and flood hazard impact assessments, to quantify population access to health services and inform strategic health system planning.
- Lead capacity strengthening initiatives, including training and mentoring researchers and programme staff in geospatial analysis, data visualization, and data management, empowering partners to independently analyze and model their own health data.
- Currently developing research on flood-induced disruption to health facility access in Tanzania, integrating JRC GloFAS flood hazard data, OpenStreetMap road networks, and travel-time modelling across 11,880 essential health facilities, with findings accepted for oral presentation at the 21st International Congress on Infectious Diseases (ICID) 2026, ASTMH 2026 and NIMR-AJSC 2026.

Data Analyst Consultant 2024 to 2025
Weill Cornell Medicine / Catholic University of Health and Allied Sciences (CUHAS) Mwanza, Tanzania

- Provided data analysis and management support for a cross-sectional clinical study examining respiratory dysfunction in people with HIV (PWH) and people without HIV (PWoH) with a history of tuberculosis in Tanzania (Respiratory Dysfunction and Self-Reported Tuberculosis History Study, Mwanza HIV & CVD Cohort).
- Managed and analyzed spirometry data collected using EasyOne Air to American Thoracic Society/European Respiratory Society standards, and respiratory quality-of-life data using the validated Kiswahili St. George's Respiratory Questionnaire (SGRQ).
- Conducted statistical analyses to characterize the primary composite outcome, any respiratory dysfunction, across the longitudinal Mwanza HIV & CVD cohort enrolled from October 2024 to April 2025, including univariate and multivariable logistic regression to identify risk factors for airway obstruction and restriction, linear regression, interaction/effect-modification analysis, and imputation of missing data across merged multi-source datasets.
- Ensured data quality and compliance with ethics approvals from Weill Cornell Institutional Review Board (IRB) and Tanzania's National Institute for Medical Research (NIMR).

National Data Analyst, WHO E-MOTIVE Project November 2020 to May 2023

Muhimbili University of Health and Allied Sciences (MUHAS) Dar es Salaam, Tanzania

- Managed and analyzed trial data for the E-MOTIVE postpartum haemorrhage cluster randomised controlled trial, a multi-country study spanning 80 health facilities and 330,000 women across Africa and Asia, working with antenatal care data and building direct experience in Reproductive and Child Health (RCH).
- Developed an interactive REDCap-based dashboard tool for tracking WHO E-MOTIVE bundle utilization across intervention facilities, improving data monitoring efficiency by 15%.
- Established a Standard Operating Procedure (SOP) for data handling adopted by five countries conducting the same study, achieving a 20% increase in data quality and consistency.
- Configured REDCap on study tablets for real-time data collection, reducing data collection time by 25%.
- Trained and mentored 56 research nurses in database best practices, exceeding implementation benchmarks by 10%.
- Compiled monthly and quarterly data reports exceeding stakeholder and donor expectations for timeliness and accuracy.
- Conducted routine data quality assessments and audits against agreed-upon indicators.

Research Scientist Intern (concurrent) November 2021 to December 2022

Ifakara Health Institute, SHLC Project Dar es Salaam, Tanzania

- Collaborated on scientific activities within the Sustainable Healthy Learning Cities and Neighbourhoods (SHLC) project, improving data gathering efficiency and reporting accuracy by 20%.
- Developed Tableau visualizations for project presentations, improving data visualization effectiveness across the team.
- Conducted robust data analysis using R and STATA, achieving a 25% enhancement in trend analysis of NCDs across diverse regions.
- Authored project reports and manuscripts for publication, contributing to a 20% increase in publication output; disseminated findings to key partners and policymakers.

Data Manager September 2015 to March 2020

Mwanza Intervention Trials Unit (MITU) Mwanza, Tanzania

- Served as Data Manager across approximately five concurrent studies spanning experimental, observational, cross-sectional, and longitudinal designs, overseeing data completeness and accuracy and achieving 98% data integrity through study-specific validation scripts in STATA and R.
- Configured GPS devices and managed household geolocation data collection for a study, followed by spatial cleaning, mapping and exploratory analysis of the geopoints.
- Designed and developed a scheduling tool to track lost-to-follow-up subjects in clinical trials, retaining over 97% of participants.
- Built, configured, and tested study databases and data entry screens using OpenClinica, REDCap, and ODK, accelerating data entry and enabling rapid analysis.
- Developed data collection tools and built automated data-cleaning systems that flagged inconsistencies for review; developed Standard Operating Procedures (SOPs) for data management across studies.
- Conducted statistical analyses including linear regression and imputation of missing data, e.g. a longitudinal analysis of BMI change following dolutegravir (DTG) initiation in HIV/hypertension patients using merged enrolment, follow-up, and clinical datasets, and prepared visualizations for publication-ready outputs, supporting multiple peer-reviewed publications.
- Automated database backup scripts, reducing backup time by 85%.
- Trained and built capacity of data clerks, data managers, and research scientists on database usage and analytical tools.

Tutor May 2012 to September 2015

University of Dar es Salaam Computing Centre Dar es Salaam, Tanzania

- Delivered lectures in Computing Mathematics I & II, Programming in Visual Basic, and System Analysis and Design, achieving a 20% average improvement in student performance.
- Developed and delivered customized professional courses on CTC database management for external organizations.
- Supervised student projects and supported research proposal writing and system development.
- Reviewed and updated study materials regularly to maintain relevance and effectiveness.

EDUCATION

Master of Science, Health Informatics — Completed 2025
University of Dar es Salaam, Tanzania

Bachelor of Science, Information Systems — Completed November 2011
University of Dodoma, Tanzania

TECHNICAL SKILLS

Programming	R (advanced), Python, STATA, SQL, Visual Basic
Geospatial	QGIS, terra, sf, Google Earth Engine, travel-time modelling, GloFAS flood hazard analysis
Data Engineering	ETL pipeline design, DuckDB, GeoParquet, GitHub Actions, REDCap, OpenClinica, ODK, data harmonization
Visualization	Shiny, Quarto, Tableau, ggplot2, interactive dashboards
Research Methods	Epidemiological analysis, health facility accessibility modelling, spirometry data analysis
Capacity Building	Researcher training, SOP development, data quality assurance, mentorship

TRAINING FACILITATED

- Trainer/Facilitator, AMMnet Tanzania Chapter "Introduction to R Training" (October 2026): taught reproducible workflows with R Markdown to about 30 students, lecturers and public health specialists; awarded a Certificate of Appreciation.
- Facilitated the AMMnet Hackathon "Building Interactive Malaria Dashboards Using R Shiny" (online, July 2026), engaging 140+ participants globally.
- Co-delivered the MAP Early Career Researcher (ECR) Training and Mentorship Programme, a six-month series of 18 hybrid workshops, networking sessions, and group projects strengthening malaria-focused research capacity across Africa (10 participants); teach data management and functional programming in R.
- Delivered geospatial analysis training to national malaria programmes and partners: Zanzibar Malaria Elimination Programme (ZAMEP), Malaria Elimination in Zanzibar (MEZA), and the State University of Zanzibar (SUZA), training 30+ participants.

PROFESSIONAL SERVICE

- Event Moderator, 2nd International Biometric Society (IBS) Tanzania Region Symposium, "Application of AI in Healthcare Research" (virtual, 19 to 20 November 2026).

SHORT COURSES ATTENDED

Geospatial Modelling for Public Health Applications, UDSM Statistics & IHI	September 2025
Advanced Course in Epidemiological Analysis, LSHTM	September 2023
Survey Methodology and Data Management, GESIS Leibniz Institute / MRC Uganda	October 2022
Introduction to Data Analysis Using R, GESIS Leibniz Institute / MRC Uganda	October 2022
Leadership and Management Skills for Healthcare Professionals, Univ. of Birmingham (Online)	July 2022
Health Data Analytics for COVID-19 Response, UWC / Hochschule Neu-Ulm (Online)	November 2021
Using STATA for Health Research, MRC/UVRI & LSHTM Uganda	November 2019
Certificate in Research Methodology, NIMR & MITU Tanzania	March 2019

CONFERENCES AND WORKSHOPS

- American Society of Tropical Medicine and Hygiene (ASTMH) Annual Meeting, Canada, November 2026 (upcoming).
- 2nd Primary Health Care Congress, Amref International University (AMIU) Main Campus, Nairobi, Kenya, March 2026.
- NIMR Annual Joint Scientific Conference (AJSC), AICC, Arusha, Tanzania, 2026: presented work on flood exposure and health service access.

AWARDS

- HIV Research Trust Scholarship: awarded £10,000 to undertake the Advanced Course in Epidemiological Analysis at the London School of Hygiene & Tropical Medicine (LSHTM).

- International Summer School in Uganda (ISSU) Scholarship: selected for a fully funded place on the 2nd International Summer School in Uganda on Survey Methodology and Data Management.

PROFESSIONAL MEMBERSHIPS

- Applied Malaria Modeling Network (AMMnet), member and technical workgroup participant
- International Biometric Society (IBS), Tanzania Region
- American Society of Tropical Medicine and Hygiene (ASTMH)

PUBLICATIONS & MANUSCRIPTS

Peer-reviewed articles

1. Manyangu GJ, Wajanga B, Desderius B, Fadhili S, Willkens M, **Myalla C**, Kisigo GA, Praygod G, Lee MH, Peck RN, Kalluvya S, Yan LD. Effect of dolutegravir on "Life's Simple 7" cardiovascular risk factors: a longitudinal analysis from Tanzania. *HIV Medicine*. 2026. doi:10.1111/hiv.70267. PMID: 42289284; PMCID: PMC13358882.
2. Reis KG, Desderius B, Kingery J, Kirabo A, Makubi A, **Myalla C**, Lee MH, Kapiga S, Peck RN. Blood pressure, T cells, and mortality in people with HIV in Tanzania during the first 2 years of antiretroviral therapy. *Journal of Clinical Hypertension*. 2020;22(9):1554-1562. doi:10.1111/jch.13975. PMID: 32815636; PMCID: PMC7722134.

Under review

3. Kreniske J, Ruselu G, Kaji A, Samwel H, **Myalla C**, Liseki E, Kwiolecha E, Desderius B, Lee MH, Zifodya JS, Kaner RJ, Fitzgerald DW, Peck RN. COPD in Tanzanian never smokers with and without HIV. Brief report, under review.

Conference presentations

4. **Myalla C**, Kafwimi N, Kafwanka P, Diao O, Vargas C, Van Den Berg M, Ketende S, Amratia P, Rumisha S. Flood exposure and disruption of geographic access to essential health services in Tanzania. Accepted for oral presentation: 21st International Congress on Infectious Diseases (ICID), Madrid, Spain, 2026; ASTMH Annual Meeting 2026; NIMR Annual Joint Scientific Conference (AJSC) 2026; 2nd Primary Health Care Congress 2026.

REFEREES

Available on request.